Information & Management 50 (2013) 489–506

Contents lists available at ScienceDirect

Information & Management

journal homepage: www.elsevier.com/locate/im

Case studies in research

Understanding the impact of business cases on IT investment decisions: An analysis of municipal e-government projects

Egon Berghout [a,*], Chee-Wee Tan [b]

[a] Department of Innovation Management & Strategy, Faculty of Economics and Business, University of Groningen, The Netherlands
[b] Department of IT Management, Copenhagen Business School, Denmark

ARTICLE INFO

Article history:
Received 18 June 2012
Received in revised form 1 July 2013
Accepted 24 July 2013
Available online 1 August 2013

Keywords:
Business case
Decision making
Information systems
Information management
IT assessment
IT evaluation
Economies of scale

ABSTRACT

This study synthesizes the extant literature to derive an integrative developmental framework for IT business cases that can be applied to diagnose the feasibility of technological investments. We then construct a theoretical model that postulates the impact of IT business case elements on the initial cost estimates of technological investments. Subsequently, our theoretical model is subjected to empirical validation through content analysis of IT business cases developed for municipal e-government projects. Findings indicate that the richness of the richness of business cases translates to more initial costs being identified in technological investments, thereby conserving resources for the organization through informed investment decisions.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

While IT investments consume 20–40% of organizations' overall IT expenditures [55], the failure rates of such investments are infamous. The 'CHAOS' report, released by the Standish Group [70], indicated that 44% of IT projects exceeded their initial budget estimates while 24% were abandoned prior to completion or never delivered. Keil et al. [34] also painted a bleak picture by confirming that escalation occurs in 38% of IT projects with 82% of these exceeding their original budgets (see also [19,58]).

Fortune and White [23], in a review of the extant literature on project management, identified the development of business cases as a key criterion for project success. The same can be said for IT investments. Kappelman et al.'s [32] survey of 55 project managers revealed that the absence of a business case constitutes an 'early warning sign' of failed IT projects (see [80]). Ward et al. [81] further alleged that the richness of business cases bolsters the success of IT investments by offering a complete breakdown of cost elements that informs investment decisions (see also [20,57,62,76,84]). Indeed, Whittaker [83] demonstrated that incomplete business cases erode the success of IT investments by culminating in budget overruns due to a priori cost underestimation.

Business cases assess multiple dimensions of IT projects, including the technology being deployed, its impact on targeted users and its implications for corporate strategies and business processes. Business cases hence determine the success of IT investments by empowering organizations to: (1) undertake informed decisions regarding IT projects; (2) monitor the progress of invested projects; and (3) evaluate project outcomes upon completion [4,62,81]. Business cases should lead to improvements in project governance and, in turn, decrease the failure rates of IT investments [80]. Yet, despite the importance of business cases, empirical investigations of business case development methodologies remain scant.

This study therefore contributes to an in-depth understanding of business case development for IT projects by deriving the set of constituent elements that facilitate an informative and purposeful approach to investment decisions. Specifically, we endeavor to answer the following research question: "*What are the constituent elements of IT business cases, and do these elements add value to IT investment decisions?*" In answering the aforementioned research question, we contribute to the extant literature in two ways. First, we identify constituent elements that support the development of rich business cases for IT investments. Second, we determine the extent to which these business case elements inform IT investment decisions.

We begin by synthesizing the extant literature on project evaluation to elicit and consolidate the constituent elements of IT business cases. Next, we construct a theoretical model that

* Corresponding author at: Department of Innovation Management & Strategy, Faculty of Economics and Business, University of Groningen, P.O. Box 800, 9700 AV, The Netherlands. Tel.: +31 50 363 3721; fax: +31 50 363 2275.
E-mail address: e.w.berghout@rug.nl (E. Berghout).

0378-7206/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.im.2013.07.010

postulates the impact of these constituent elements on IT investment decisions, specifically with respect to cost estimations. This model is then subjected to empirical validation through a content analysis that examines (1) the business cases of comparable e-government projects for municipalities and (2) their impact on the initial cost estimates of these projects. Given the paucity of studies exploring the value of IT business cases, the results of this investigation make a novel contribution to theory by (1) deriving a framework for developing IT business cases that delineates constituent case elements, and; (2) determining the marginal value of each case element in informing investment decisions.

This paper comprises four sections, including Section 1. In Section 2, we compare and contrast contemporary business case development methodologies to advance an integrative framework for developing IT business cases. Based on this framework, a theoretical model and testable hypotheses are developed to predict the impact of different business case elements on the initial cost estimates of IT investments. Next, Section 3 elaborates on the data collection and analytical procedures that were employed to test the applicability of our proposed developmental framework and validate the hypotheses in our theoretical model. Finally, Section 4 discusses our empirical findings and their implications for both theory and practice. This section also highlights the limitations of our study and suggests potential avenues for future research.

2. Theoretical foundation

As organizations struggle to realize the benefits of IT investments, repeated calls for a comprehensive and robust framework to evaluate the feasibility of such investments have been made by both scholars and practitioners alike (e.g., [56,70,81,84]. Yet, as noted by Melville et al. [48], the extant literature on IT evaluation is fragmented, as past studies have advocated a variety of evaluation criteria. These myriad evaluation criteria in turn, can be broadly classified into two schools of thought, namely the *Economic Approach* and the *Socio-Technical Approach*. Whereas the *Economic Approach*seeks to quantify the costs and benefits of IT investments in terms of financial metrics [14,43], the *Social-Technical Approach* regards IT investments as a form of organizational intervention that embodies both social and technical elements of change [71,77]. Although the economic approach has dominated prior research in IT evaluation [24], it suffers from two distinct drawbacks: (1) an overemphasis on potential revenue growth rather than cost displacement [21] and, (2) long-standing challenges faced in cost–benefit analysis given the tangible and intangible impacts of technologies [12,22]. Indeed, there is growing evidence to suggest that the socio-technical approach circumvents the limitations of the economic approach by taking into account both quantifiable and non-quantifiable aspects of IT investments [36,39,41,47]. For instance, Ward et al.'s [82] survey of 100 European organizations concluded that organizations with greater success in IT investments are often those subscribing to a broader definition of benefits such as innovation and enhanced cooperation with trading partners (see also [72]). In contrast, less successful organizations tend to restrict their evaluation of IT investment benefits to those associated with business performance and efficiency improvements [82].

A review of published articles on IT project evaluation, which have appeared in leading journals[1] for the last five years (i.e., 2007–2012), painted an identical picture (see Appendix B). While a fraction of these studies persisted in applying an economic approach to the evaluation of IT projects [1,16,29,35,49,67,68,74,78], the bulk of them tend to rely on evaluation criteria that recognize both quantifiable and non-quantifiable aspects of IT projects [3,5,7–9,13,18,26,28,36,38,40,42,47,54,60,69,73,79,85]. Consistent with an increasing emphasis on the socio-technical approach to IT evaluation, the development of business cases has emerged as a standard practice among organizations when undertaking IT investments [81]. As defined by Matejka and Cossé [46], a business case is a "*description of a situation or sequence of events confronting an individual, a set of individuals, or an organization [and] includes a detailed account of the events leading to the point in time at which the case concludes*". When applied to IT investments, business cases capture the rationale behind IT projects and in so doing, safeguard against resource wastages by ensuring that project outcomes satisfy important business motives [10,11,15]. Ward et al. [81] thus claimed that the development of business cases is an essential step in IT evaluation because it (1) aids in the prioritization of competing investment options based on resource constraints; (2) facilitates the formulation of concrete plans to realize benefits from investments; (3) elicits commitment from stakeholders to attain projected investment benefits; and (4) creates a formal basis for reviewing the eventual outcomes of these investments. Nevertheless, while scholars have generally acknowledged to the value of business cases in bolstering the success of IT investments (e.g., [20,32,57,62,76,80–84]), few studies clarify how such business cases can be developed comprehensively. This study therefore contributes to the literature by synthesizing contemporary business case methodologies to derive a developmental framework for business cases that allows organizations to systematically reflect on the viability of IT investments.

2.1. Business case methodologies: an overview

Although research on IT evaluation is gaining momentum, studies on business cases remain scarce. To solicit business case methodologies from the extant literature, we began with a keyword search of 500,000 periodicals and 12 million books from the EBSCO database, which includes Business Source Premier and Econlit, yielding 254 articles in the process. A similar search conducted via Google (Scholar) generated an additional 10 articles, resulting in a total of 264 articles for review. Going through the abstracts of these 264 articles, we arrived at 89 articles that (directly or indirectly) touch on the evaluation of IT investments. Crosschecking the author(s) and references of these 89 articles, we added another 30 articles to the pool, resulting in a final sample of 119 articles for analysis. Appendix A provides a schematic overview of the aforementioned literature review process.

Further analysis of these 119 articles reveals that a majority relates to IT justification in general, and we can only uncover eight prominent business case methodologies for evaluating IT investments. From these eight methodologies, we endeavor to elicit and consolidate elements that constitute an integrated developmental framework for IT business cases. Such an approach precludes the omission of business case elements that are paramount to the evaluation of IT investments. In subsequent sections, we introduce and compare these eight business case methodologies in constructing our developmental framework.

2.1.1. Barnes's business case methodology

Barnes [4] defined a business case as "an internal proposal that is submitted to management for a decision" (p. 12). A well-developed business case must therefore provide answers to four fundamental questions: the extent to which (1) an organization is in need of the projected benefits from an IT investment; (2) the

[1] Journals reviewed include the *European Journal of Information Systems* (EJIS), *Information Systems Journal* (ISJ), *Information Systems Research* (ISR), *Information & Management* (I&M), *Journal of the Association of Information Systems* (JAIS), *Journal of Information Technology* (JIT), *Journal of Management Information Systems* (JMIS), *Journal of Strategic Information Systems* (JSIS) and *MIS Quarterly* (MISQ).

Table 1
Constituent dimensions of barnes' business case methodology [4].

Constituent dimension	Definition
Management summary	Condensed overview of the intended IT investment by laying out key objectives, issues and timetables
Business needs	Description of the business goals behind the intended IT investment
Requirements	Clarification of the scope of the intended IT investment including its functional, technical and operational requirements to meet targeted business goals
Supply options	Identification of potential suppliers who can best satisfy business requirements
Costs	Estimation of one-off (initial) (e.g., communication equipment, hardware installations, software licenses and legal fees), recurring (running) (e.g., support and maintenance of both hardware and software) and other related costs (e.g., personnel and training)
Benefits	Projection of tangible financial benefits to be obtained from the intended IT investment
Risk analysis	Assessment of impacts, uncertainties and other consequences associated with the intended IT investment

organization possesses the capability to realize those projected benefits; (3) the investment is risky in nature; and (4) projected benefits are being attained at optimum cost [4]. To provide answers to the aforementioned questions, Barnes [4] recommended that the development of a business case should incorporate seven key dimensions as summarized in Table 1.

2.1.2. Remenyi's business case methodology

Remenyi [62] defined a business case as a "justification for pursuing a course of action in organizational context to meet stated organizational objectives or goals" (p. 6). In this sense, a well-developed business case stresses an alignment between the strategic targets of an organization and its intended IT investment. According to Remenyi [62], such business cases typically comprise five key dimensions as summarized in Table 2.

2.1.3. Cresswell et al.'s business case methodology

As argued by Cresswell et al. [17], a business case represents "a well-reasoned argument that attempts to convince the audience of the benefits of investment while educating them about the changes, costs and risks that will be part of the effort" (p. 25). Consequently, Cresswell et al. [17] put advance six business case guidelines that touch on areas that are likely to be of interest to organizations undertaking IT investments (see Table 3).

2.1.4. Schmidt's business case methodology

Schmidt [65] maintained that the basic objective of a business case is to shape the decision making process by bringing to light various organizational concerns related to the intended IT investment. Because IT investments denote intervention strategies undertaken by organizations to improve existing business conditions, Schmidt [65] proposed five checkpoints for business cases that attest to the effectiveness of the intervention. These checkpoints denote key aspects of IT investments that distinguish a baseline 'As-Is' scenario from projected 'To-Be' scenarios for organizations (see Table 4).

2.1.5. The Australian government's business case methodology

In June 2008, the Australian Government initiated a program to develop business case guidelines for governmental organizations embarking on IT investment projects exceeding AUD $10 million. Based on the format introduced by the Australian Government [2], a business case supplies valuable information on the costs, benefits and risks associated with an intended IT investment and consists of eight key dimensions as summarized in Table 5.

2.1.6. Frisk's business case methodology

Scrutinizing the way IT investments are evaluated within non-profit organizations, Frisk [24] observed that business cases are too often interpreted as financial instruments rather than as value propositions regarding intended investments. Frisk [24] described the business case as a formative and pluralistic evaluation method that reflects decision makers' projections of how the intended IT investment culminates in improved efficiency (i.e., doing things right) and effectiveness (i.e., doing the right things) for the organization. As value propositions, Frisk [24] claimed that a business case should include the eight dimensions summarized in Table 6.

2.1.7. Gambles' business case methodology

Gambles [25] defined a business case as "a recommendation to decision makers to take a particular course of action for the organization, supported by an analysis of its benefits, costs and risks compared to the realistic alternatives, with an explanation of how it can best be implemented" (p. 1). Gambles [25] introduced five business case dimensions that assist decision makers in planning and directing IT investments (see Table 7).

2.1.8. The Scottish government's business case methodology

The Scottish Government [66] presented five business case guidelines as a means of improving the probability of beneficial investment decisions. These guidelines cater to both financial and non-financial facets of IT investments as summarized in Table 8.

2.1.9. Toward an integrative developmental framework for IT business cases

Comparisons among the eight aforementioned business case methodologies yield valuable insights to guide the construction of an integrative developmental framework for business cases (see Table 9). Not surprisingly, all eight business case methodologies recommended the inclusion of '*business objectives*', '*costs appraisal*', '*benefits appraisal*' and '*risk assessment*', thereby attesting to their significance in the evaluation of IT investments. Of the eight

Table 2
Constituent dimensions of Remenyi's business case methodology [62].

Constituent dimension	Definition
Business outcome	Specification of the desired effect of an intervention or change to a business process or practice brought about by the intended IT investment
Stakeholders	Identification of all parties involved in the intended IT investment and clarification of requirements for information exchange and collaboration among them
Strategic Alignment	Description of how the intended IT investment will fulfill business goals
Technology	Determination of the suitability of the technology to be invested in and the feasibility of the organization's implementation plans
Risks	Assessment of business, developmental and architectural uncertainties associated with the intended IT investment

Table 3
Constituent dimensions of Cresswell et al. business case methodology [17].

Constituent dimension	Definition
Problem statement and objectives	Description of existing problems faced by the organization and the business goals to be attained to resolve them
Approach and alternatives	Specification of opportunities for intended IT investments and selection of a viable approach to achieve targeted business goals
Stakeholders	Identification of all parties involved in the intended IT investment and clarification of key concerns and avenues for possible collaboration
Benefits and costs	Estimations of implementation and maintenance costs of the intended IT investment and individual and collective benefits for key stakeholders
Risks	Assessment of uncertainties associated with the intended IT investment together with suggestions for how to mitigate such uncertainties
Project planning	Determination of milestones, resource capacity and timelines for the intended IT investment

Table 4
Constituent dimensions of Schmidt's business case methodology [65].

Constituent dimension	Definition
Introduction and overview	Specification of the business goals to be attained from the intended IT investment and outlining a viable course of action to evaluate the impact on the organization
Assumptions and methods	Description of pre-conceptions related to the intended IT investment and explanation of the methods utilized to estimate its financial costs
Business impacts	Projection regarding economic returns from the intended IT investment
Sensitivity, risks and contingencies	Assessment of uncertainties associated with the intended IT investment by: (1) mapping out the spectrum of possible deviations from initial projections; (2) specifying the risks to be borne in the investment given the possibility of such deviations; and (3) proposing remedial actions that could be taken to mitigate these risks
Conclusions and recommendations	Determination of the feasibility of the intended IT investment by: (1) weighing its financial costs against attainable business benefits; and (2) assigning roles and responsibilities to stakeholders in a manner that contributes to the success of the investment

Table 5
Constituent dimensions of the Australian government's business case methodology [1].

Constituent dimension	Definition
Objectives and approach	Condensed overview of the intended IT investment by laying out priorities, benefits, costs and risk pressures
Preparation and strategic alignment	Description of the assumptions and rationale behind the intended IT investment together with clarification on how the investment aligns with governmental policies and service delivery objectives
Demand, benefits and IT gap analysis	Identification of demand sources and end-user characteristics that warrant attention and the formulation of a benefits realization plan including key performance indicators (KPIs)
IT options analysis	Determination of technological shortcomings to be addressed via the intended IT investment to meet targeted business goals
Schedules and governance	Specification of timeline for the intended IT investment and identification of governance mechanisms to cope with project planning and stakeholder interactions
Risk identification and mitigation	Assessment of uncertainties associated with the intended IT investment together with an exit strategy in case the investment turns out to be unfavorable
Cost estimates	Estimation of cost and economic viability of the intended IT investment
Consolidation	Summary of the business case, including approval signatures from designated decision makers

Table 6
Constituent dimensions of Frisk's business case methodology [24].

Constituent dimension	Definition
Strategic match	Description of how the intended IT investment matches targeted business goals
Impact on the surroundings	Specification of how the intended IT investment affects trading partners of the organization
Organizational impact	Specification of how the intended IT investment affects the organization
Benefits and costs	Estimations of implementation and maintenance costs for the intended IT investment as well as individual and collective benefits for key stakeholders
Risks	Assessment of uncertainties associated with the business aspects of the intended IT investment (e.g., business continuity)
IT systems and technology	Assessment of uncertainties associated with the technological aspects of the intended IT investment (e.g., information security and IT architectural fit)
Stakeholders' opinion	Identification of all parties involved in the intended IT investment and clarification of key concerns associated with the intended IT investment
Project organizing	Determination of milestones, resource capacity and timelines for the intended IT investment

Table 7
Constituent dimensions of Gambles' business case methodology [25].

Constituent dimension	Definition
Strategic Fit	Specification of how the intended IT investment matches targeted business goals
Stakeholder assessment	Identification of parties that possess both high interest in the intended IT investment and an ability to influence the outcomes of the investment
Benefits mapping	Projection of both tangible and intangible benefits to be obtained from the intended IT investment
Costs modeling	Estimation of implementation and maintenance costs of the intended IT investment
Risk analysis	Assessment of uncertainties associated with the intended IT investment

Table 8
Constituent dimensions of the Scottish government's business case methodology [66].

Constituent dimension	Definition
Strategic case	Determination of objectives, existing arrangements and business goals of the organization, as well as specification of business scope and key service requirements for the intended IT investment
Procurement strategy	Identification of potential suppliers who can best satisfy business requirements and explanations of contractual negotiations and risks associated with these suppliers
Economic appraisal	Estimation of implementation and maintenance costs of the intended IT investment
Benefits appraisal	Description of benefit realization plan to realize both tangible and intangible benefits from the intended IT investment
Risk assessment and sensitivity analysis	Assessment of uncertainties associated with the intended IT investment together with suggestions for mitigation of such uncertainties

business case methodologies, six recommended the inclusion of '*stakeholders*' to clarify the degree of involvement for different stakeholders and allocate responsibilities in a manner that contributes to the success of the intended IT investment. Five business case methodologies recommended the inclusion of '*technological requirements*' to indicate how the intended IT investment fulfills the business goals of the organization. Four business case methodologies recommended the inclusion of '*project planning and governance*' where concrete milestones and timelines for the IT investment are laid out. Only two business case methodologies recommended the inclusion of '*supplier options*' to highlight the importance of considering multiple vendors to select the best alternative for the IT investment. Likewise, two business case methodologies recommended the inclusion of '*consolidation*' to summarize key agreements on the IT investment and assign responsibilities to all involved parties.

Though there are considerable differences among the eight business case methodologies, our comparison of these methodologies reveals nine constituent dimensions that are pertinent to the evaluation of IT investments (see Table 9). Indeed, most if not all of these nine dimensions coincide with core factors identified in IT justification studies (e.g., [33,62]) and overlap with evaluation criteria acknowledged in past studies of IT projects from our literature review (see Appendix B and Table 10). We hence posit that the preceding nine dimensions constitute an integrative developmental framework for business cases.

2.2. Theoretical model of the impact of business case constituents on initial cost estimates of IT investments

Exceeding initial budget projections remains an elusive and persistent challenge confronting organizations when undertaking IT investments [34,70,86]. Whenever IT investments exceed their initial budget projections, organizations may be compelled to make hasty decisions with respect to project continuation and/or cancelation [51], thereby leading to situations of unwarranted escalation or premature abandonment. Indeed, as observed by El Emam and Koru [19] in a longitudinal survey of IT project managers, going over budget has been cited as one of the main reasons behind project cancelations (see also [1,16,29]). These sentiments were echoed by Procaccino et al. [59], who noted that disappointment with IT projects is more often caused by expectations established at the outset than events that occur afterwards (see also [81]). A specification of cost estimates before embarking on IT investments would therefore reduce if not eliminate the problem of budget overruns and conserve organizational resources that could otherwise be wasted on unwarranted escalation or premature abandonment decisions. Consequently, this study examines the impact of business case elements on cost estimates for IT investments such that investment decisions can be undertaken in an informed manner.

Remenyi [62] contended that the richness of business cases is pivotal to the evaluation of IT investments. As maintained by Money et al. [50], the development of a business case constitutes an extensive and structured approach to evaluating the intended IT investment in its multi-dimensional facets. That is, a business case combines both financial (e.g., cost estimations and productivity improvements) and non-financial (e.g., stakeholder involvement and strategic alignment) aspects of IT investments into a blueprint that charts out the best course of action for organizations [65]. Conceivably, the more elements addressed in a business case, the more likely it is able to identify investment costs upfront and prevent hidden costs from surfacing much later on in the project. We therefore hypothesize that:

Hypothesis 1. The number of elements in a business case is positively related to the initial cost estimates of an IT investment.

Yetton et al. [86] alleged that the *organizational*, *technical* and *project* characteristics of an IT investment determine of its performance (see also [63,64]). As stated by Yetton et al. [86], the organizational characteristics of an IT investment govern management's commitment to and support for the investment and include factors such as well-articulated business objectives and proper communication. Conversely, technical characteristics concern the technological novelty of an IT investment and as argued by Yetton et al. [86], may influence investment performance whenever the investment revolves around the implementation of new and innovative technologies. Finally, project characteristics are predictive of the performance of an IT investment because they reflect the extent to which conditions conducive to project success are present: these conditions include sufficient budgets and the involvement of stakeholders with the required knowledge and technical expertise [86]. Consistent with Yetton et al.'s [86] IT project performance model, we classify the nine business case elements in our developmental framework into categories of *organizational*, *technological* and *project* constituents to further postulate the effects of these business case elements on the cost estimates of IT investments.

2.2.1. Organizational constituents

A business case should clearly describe the current situation and targeted business objectives to be realized via the intended IT investment [81]. Larson and Gobeli [37] contended that higher chances of success are often derived from investments with well-defined business objectives. Byrd et al. [15] also demonstrated that higher returns from IT investments are likely to result from better business-IT alignment (see also [27,60,61]). In contrast, weak alignment between business and technology will mislead organizations into delaying or even abandoning IT investments, that are competitively necessary [52]. In addition, Neirotti and Paolucci [52] recommended that organizations consolidate IT investments (whenever possible) to: (1) remove redundant business requirements; (2) initiate innovation; and (3) avoid delays in the exploitation of emerging technologies. As one of the critical success factors behind successful IT projects, consolidation can reduce maintenance costs from legacy systems and allow the channeling of resources to other urgent forms of IT investment [52]. From the

Table 9
Comparison of contemporary business case methodologies.

Barnes [4]	Remenyi [62]	Cresswell et al. [17]	Schmidt [65]	Australian Government [2]	Frisk [24]	Gambles [25]	Scottish Government [66]	Proposed Developmental Framework
Management summary/ business needs	Business outcomes/ strategic alignment	Problem statement and objectives	Introduction and overview	Objectives and approach	Strategic match/ organizational impact	Strategic fit	Strategic case	Business case objectives
Requirements	Technology	Approach and alternatives	No comparable element	IT gap analysis/IT options analysis	IT systems and technology	No comparable element	No comparable element	Technological requirements
Supply options	No comparable element	No comparable element		No comparable element	No comparable element		Procurement strategy	Supplier options
Costs	Business outcomes	Benefits and costs	Business impacts	Cost estimates	Benefits and costs	Costs modeling	Economic appraisal	Costs appraisal
Benefits			Assumptions and methods	Preparation and strategic alignment/demand and benefits		Benefits mapping	Benefits appraisal	Benefits appraisal
No comparable element	Technology/business outcomes	Project planning	No comparable element	Schedules and governance	Project organizing	No comparable element	No comparable element	Project planning and governance
Risk analysis	Risk	Risks	Sensitivity, risks & contingencies	Risk identification & mitigation	Risks	Risk analysis	Risk assessment and sensitivity analysis	Risk assessment
No comparable element	Stakeholders	Stakeholders	Conclusions and recommendations	Schedules and governance	Impact on the surroundings/ stakeholders' opinions	Stakeholder assessment	No comparable element	Stakeholders
	No comparable element	No comparable element		Consolidation	No comparable element	No comparable element		Consolidation

above, we can thus surmise that 'Business Case Objectives', 'Benefits Appraisal' and 'Consolidation' in a business case serve as *Organizational Constituents* that govern how organizations can go about realizing their targeted business objectives through IT investments. Further, we hypothesize that the greater the number of organizational constituents addressed in a business case, the higher are the cost estimates for IT investments because they advance a more holistic view of what the organization lacks:

Hypothesis 2. The number of organizational constituents in a business case is positively related to the initial costs estimates of an IT investment.

2.2.2. Technological constituents

In examining unsuccessful IT projects across organizations with of varying sizes and situated within different industries, Whittaker [83] found that the adoption of novel (or unproven) technologies with poorly specified requirements (see also [7,8,28,67,68,79]) and the selection of vendors that are incapable of completing projects in a timely fashion and according to predetermined specifications represent key factors contributing to project failures (see also [23]). Similar conclusions were reached by El Emam and Koru [19], who observed that investing in unproven technologies is a primary reason behind project cancelations, especially when it culminates in ever-changing technological requirements and project scope. Because a thorough analysis of 'Technological Requirements' and 'Supplier Options' translates into higher investment costs due to the accommodation of a larger set of technical considerations, we hypothesize that the greater the number of *Technological Constituents* (i.e., elements concerning the acquisition of appropriate technological solutions) being addressed in a business case, the higher the cost estimates of IT investments:

Hypothesis 3. The number of technological constituents in a business case is positively related to the initial costs estimates of an IT investment.

2.2.3. Project constituents

Another oft-cited reason for IT project failures is inadequate risk management [53,73,83]. Poor risk analysis and management results in serious budget overruns due to a priori cost underestimation [83]. By improving the ability to identify and manage risks associated with IT investments, organizations can remain vigilant against deviations from expected outcomes and avoid project failures [31,85]. Apart from risk assessment, other reasons for IT project failures – as uncovered through the work of Whittaker [83] and subsequently confirmed by Fortune and White [23] as critical determinants of the success of IT investments – include flawed cost estimations, insufficient project planning and governance and a lack of stakeholder involvement and support (see also [18,38,42,47]). Together, these aforementioned elements (i.e., project planning and governance, cost estimates, risk assessment and stakeholders) capture project-specific concerns for IT investments [6]. We therefore hypothesize that the greater the number of *Project Constituents* addressed in a business case, the higher the cost estimates of IT investments due to a more complete analysis of issues relevant to the project at hand:

Hypothesis 4. The number of project constituents in a business case is positively related to the initial costs estimates of an IT investment.

Fig. 1 illustrates our proposed theoretical model and the testable hypotheses. The next section will describe the data sample and analytical procedures that are employed to validate the model.

Table 10
Mapping of business case constituents to extant literature on IT projects [sample $N = 30$].

Author(s)	Business case objectives	Benefits appraisal	Consolidation	Technological requirements	Supplier options	Project planning & governance	Cost appraisal	Risk assessment	Stakeholders
Ashurst et al. [1]	x			x			x		
Baars et al. [3]	x	x							x
Bechor et al. [5]	x		x	x					
Benlian [7]	x	x		x	x		x	x	x
Benlian and Hess [8]	x	x	x	x	x	x	x	x	x
Berente et al. [9]						x			x
Bose et al. [13]	x	x							
Conboy [16]						x	x		
Davis et al. [18]				x					x
Grimsley and Meehan [26]		x						x	x
Häkkinen and Hilmola [28]		x	x	x	x			x	x
Henry et al. [29]						x	x		
Kesseler [35]	x	x							
Koh et al. [36]	x	x	x						x
Law and Ngai [38]				x					x
Li and Mao [40]	x	x							
Lim et al. [42]		x	x						
McNaughton et al. [47]	x	x	x	x			x		x
Mithas et al. [49]		x							
Nevo and Wade [54]	x								x
Ravishankar et al. [60]			x						
Seddon et al. [67]	x								x
Siau and Rossi [68]		x		x					
Srivastava and Teo [69]					x	x	x	x	
Taylor et al. [73]				x		x	x	x	x
Thiesse et al. [74]		x							
Wamba and Chatfield [78]		x		x					x
Wang [79]		x	x	x	x			x	x
Wu et al. [85]		x				x		x	
Total [percentage]	12 [41.4%]	17 [58.6%]	8 [27.6%]	12 [41.4%]	5 [17.2%]	7 [24.1%]	8 [27.6%]	8 [27.6%]	15 [51.7%]

3. Methodology

Our theoretical model is validated in two stages. In the first stage, content analysis was performed to: (1) elicit case elements from the business cases of 38 Dutch municipal e-government projects in accordance with our framework; and (2) investigate the extent to which different case elements are addressed in IT project planning for Dutch municipalities. We then analyzed the impact of the existence of different case elements on the initial cost estimates of these e-government projects in the second stage.

3.1. Data sample

Due to stipulations of the national e-government stimulation program, all municipalities in the Netherlands must undertake IT investments in compliance with new legislation regarding the

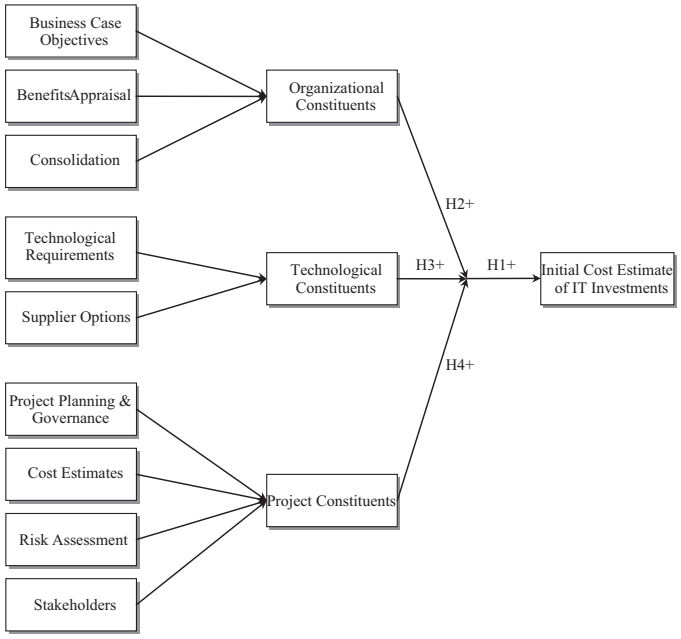


Fig. 1. Theoretical model of the impact of business case constituents on initial cost estimates of IT investments.

Table 11
Descriptive statistics of data sample.

	N	Mean	Std. deviation
Citizens	38	28,842.45	20,696.56
Total initial cost (Euro)	38	€ 808,088.58	€ 669,598.03
Initial cost per citizen (Euro)	38	€ 30.19	€ 18.02
Valid *N* (listwise)	38		

delivery of public e-services. In the process, the municipalities are encouraged to develop business cases to outline the planning behind their proposed e-government projects and justify cost estimations, although they are not obliged to do so. The contextual similarity among the municipalities' e-government projects presents a unique opportunity to compare and contrast the impact of business cases on technological investments without introducing unnecessary confounds due to project-level differences.

As part of a nation-wide program to standardize e-government services across municipalities, these projects are equally matched in terms of their strategic scope and value. Each e-government project entails the implementation of (1) predefined web features and relationship management services for firms and citizens; (2) workflow systems for the digital handling of authorizations, permissions and identification tooling; and (3) a brokerage system for connecting to information systems belonging to national agencies for the purpose of registering buildings/addresses, firms and citizens. In essence, the targeted objective of this five-year program is to bolster the effectiveness and efficiency of public e-services among municipalities. Our dataset comprises business cases of e-government projects from 38 Dutch municipalities, representing 9% of the total population of the Netherlands. These 38 business cases belong to municipalities that have formally evaluated their IT investments and attained approval from the municipal council based on these business cases. For each municipality, the dataset captures its size (i.e., number of citizens) and the estimated total initial cost for its e-government project. Table 11 summarizes the descriptive statistics of the eventual dataset.

As seen in Table 11, the high standard deviation for the initial total cost indicates that cost estimates for the e-government projects differ considerably among the municipalities. Because initial total cost estimates are likely to have a strong correlation with the size of municipalities, we computed '*Initial Cost per Citizen*' as a better gauge of cost estimates for a single municipality, as it normalizes the investment cost of an e-government project across the municipal population

3.2. 1st stage – content analysis of e-government business cases

For each business case, a content analysis was performed by two of the authors to determine whether it contains one or more of the case elements in our proposed developmental framework. Table 12 summarizes the content analysis schema utilized for this investigation.

As seen in Table 12, our content analysis schema is constructed from synthesizing previous characterizations of business case elements to guarantee internal validity and reliability in how each e-government business case is interpreted and analyzed. It should be further noted that the characterization of each of the nine business case elements remains fairly broad to ensure that empirical findings are not impacted by contextual variations among different e-government projects. For instance, we do not specify the type of costs that should be estimated for an e-government project because different municipalities might have different cost considerations. Rather, what is of interest is whether each municipality has undertaken proper cost estimations in evaluating the feasibility of its e-government project. For the same reason, each business case element is scored in a dichotomous fashion: for each of the nine business case elements, a '1′ is awarded if the element is found to be present in an e-government business case and '0′ otherwise.

Fig. 2 depicts an overview of the presence of various case elements, in descending order of frequency, as extracted from the e-government business cases of the 38 municipalities in our dataset.

Insights can be gleaned from Fig. 2 concerning the development of IT business cases for Dutch municipalities. The '*Business Case Objectives*' element can be found in the business cases of all 38 municipalities, thus implying consistency in the manner in which municipalities state the purpose of their e-government project, the business goals to be attained and the key decisions to be made.

'*Cost Appraisal*' is a core element of business cases [62,65]. While a standardized budget model is employed by 31 municipalities to estimate costs for their e-government projects, the level of detail from the remaining 7 municipalities is insufficient to arrive at proper cost estimates.

'*Project Planning & Governance*' is another prominent element found in the business cases, with 31 municipalities offering an elaborate planning and governance structure for their e-government projects. Conversely, the planning and governance structure for the e-government projects of the other 7 municipalities is too general to be informative.

The '*Benefits Appraisal*' element is found in the business cases of 27 municipalities. For these 27 municipalities, the business case contains precise guidelines for their e-government projects by explicating: (1) the exact goals to be attained and (2) the specific benefits to be realized from the investment.

Though '*Stakeholders*' are the direct beneficiaries of the e-government projects, only 21 municipalities explicitly identified the stakeholder groups to be consulted and devised ways to comply with their service expectations.

'*Technological Requirements*' are found in the business cases of 20 municipalities. The remaining 18 municipalities opted to relegate the elicitation of technological requirements to their project vendors.

Although '*Risk Assessment*' is necessary to preempt problems and secure project success, only 19 municipalities gave a thorough appraisal of risk in their business cases. From the content analysis, it appears that the other half of the municipalities do not perform risk assessments in favor of relying on the expertise of consultancies and/or the experiences of other municipalities to cope with impending problems.

The '*Supplier Options*' element concerns the evaluation of potential vendors that may be awarded the e-government projects. Yet, only 6 municipalities perform this activity whereas the remaining 32 municipalities chose to partner with familiar vendors: vendors with which the municipalities have collaborated in the past.

'*Consolidation*' is least represented in the e-government business cases, with only 4 municipalities concluding with a section where decisions are evaluated and formal agreement is established among decision makers.

3.3. 2nd stage – hypotheses testing

Regression analysis was performed to validate the hypotheses in our theoretical model and the results are summarized in Table 13. As seen in Table 13, three regression models were tested. In Model 1, we investigated the relationship between the number of business case elements and the initial cost per citizen. In Model 2, we investigated the relationship between the three constituent categories of business cases (i.e., organizational constituents,

Table 12
Constituents of business cases and their corresponding characterization for content analysis.

Constituents	Business case element	Characterization	References
Organizational – concerns the business aspects of the IT investment	Business case objectives	• Condensed overview of the business case, by identifying and explaining the purpose of the intended IT investment, targeted business objectives, proposed course of action, issues, risks and timetables • Identification of associated costs and projected benefits from the intended IT investment	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Gambles [25] Remenyi [62] Schmidt [65] Scottish Government [66]
	Benefits appraisal	• Account of tangible financial benefits that could be realized by the intended IT investment • Alignment of the intended IT investment with targeted business objectives, review of assumptions and rationale behind critical issues that impact the business	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Gambles [25] Remenyi [62] Schmidt [65] Scottish Government [66]
	Consolidation	• Overview of the business case listing business targets, financial requirements, decisions and agreed-upon actions including formal acceptance among stakeholders	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Gambles [25] Remenyi [62] Schmidt [65] Scottish Government [66]
Technological – concerns the performance aspects of the IT investment	Technological requirements	• Analysis of technical requirements to meet targeted business objectives • Compilation of demands to be met for end-users	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Remenyi [62]
	Supplier options	• Pre-selection and evaluation of potential suppliers that can meet the requirements of the intended IT investment • Assessment of probable contractual risks and options for strategic collaboration	Barnes [4] Scottish Government [66]
Project – concerns the management aspects of the IT investment	Project planning & governance	• Indication of time of delivery for the intended IT investment (e.g., milestones, schedules and timetables), as well as a clarification of the level of project coordination and governance needed	Australian Government [2] Cresswell et al. [17] Frisk [24] Remenyi [62]
	Cost appraisal	• Account of cost data and estimates for the intended IT investment, often expressed in terms of initial (e.g., hardware and legal fees), running (e.g., licenses and maintenance) and other organizational (e.g., personnel recruitment and training) costs	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Gambles [25] Remenyi [62] Schmidt [65] Scottish Government [66]
	Risk assessment	• Assessment of possible managerial and technical risks for the intended IT investment (e.g., impact on the organization's employees) including viable countermeasures (e.g., expected change in management and training procedures to be enacted for affected employees) and exit points if resolutions are not available • Generation of predictive models and alternative scenarios to cope with risks in advance whenever feasible	Australian Government [2] Barnes [4] Cresswell et al. [17] Frisk [24] Gambles [25] Remenyi [62] Schmidt [65] Scottish Government [66]
	Stakeholders	• Identification of beneficiaries and stakeholders for the intended IT investment with particular emphasis on possible interaction problems, project responsibilities and collective contributions to project success	Australian Government [2] Schmidt [65]

technological constituents and project constituents) and the initial cost per citizen. In Model 3, we investigated the relationship between each of the nine business case elements and the initial cost per citizen.

From Table 13, there is a statistically significant positive effect of the *number of IT business case elements* on *initial cost per citizen* (β = .481, t = 3.290, p = .002). Additionally, the *number of IT business case elements* explains a significant portion of the variance in *initial cost per citizen*[2] (Adjusted R^2 = .210, $F_{(1,\ 36)}$ = 10.823, p = .002) (see Table 13), thus corroborating Hypothesis 1. That is, the greater the number of IT business case elements, the higher the initial cost per citizen disclosed for a municipality's e-government project. The same trend can be observed from plotting the relationship between the initial cost per citizen and the number of IT business case elements (see Fig. 3). As illustrated in Fig. 3, the relationship between the initial cost per citizen and the number of IT business case elements has an upward slope, thereby suggesting that the inclusion of additional case elements will lead to the identification of additional costs in a municipality's e-government project.

To test Hypotheses 2, 3 and 4, we grouped case elements into the three constituent categories (i.e., organizational constituents, technological constituents and project constituents) for each municipality's business case. Regression analysis was then performed to test the relationship between the number of case

[2] Because R^2 is artificially inflated by adding predictors to the regression model, Adjusted R^2 is more suitable for comparing regression models by taking into account the number of predictors in the model.

Table 13
Results from regression analysis[a] [sample $N = 38$].

Independent variables	Model statistics					
	Model 1		Model 2		Model 3	
	Adjusted $R^2 = .210$ $[F_{(1,\ 36)} = 10.823;\ p = .002]$		Adjusted $R^2 = .204$ $[F_{(3,\ 34)} = 4.160;\ p = .013]$		Adjusted $R^2 = .155$ $[F_{(8,\ 29)} = 1.846;\ p = .109]$	
	Beta	*t* Statistics	Beta	*t* Statistics	Beta	*t* Statistics
Number of IT business case elements (out of 9)	.481**	[$t = 3.290$; $p = .002$]	–	–	–	–
Organizational constituents	–	–	.132	[$t = 0.882$; $p = .384$]	–	–
Technological constituents	–	–	.043	[$t = 0.294$; $p = .771$]	–	–
Project constituents	–	–	.472**	[$t = 3.156$; $p = .003$]	–	–
Benefits appraisal	–	–	–	–	.187	[$t = 0.995$; $p = .328$]
Consolidation	–	–	–	–	.004	[$t = 0.024$; $p = .981$]
Technological requirements	–	–	–	–	−.012	[$t = -0.077$; $p = .939$]
Supplier options	–	–	–	–	.060	[$t = 0.368$; $p = .715$]
Project planning & governance	–	–	–	–	−.057	[$t = -0.338$; $p = .738$]
Cost appraisal	–	–	–	–	.235	[$t = 1.379$; $p = .179$]
Risk assessment	–	–	–	–	.298	[$t = 1.535$; $p = .136$]
Stakeholders	–	–	–	–	.187	[$t = 0.907$; $p = .372$]

Dependent variable: initial cost per citizen.
[a] The '*Business Case Objective*' element is omitted because all government business cases contain this element and SPSS regards it as a constant.
** Beta coefficient is significant at the 0.01 level (2-tailed).

elements in each constituent category and the initial cost per citizen. Multicollinearity was not a threat in our regression model because: (1) tolerance values averaged more than .30 and (2) the maximum variance inflation factor (VIF) was well below the prescriptive diagnostic of 5.0 [27]. Together, the three constituent categories explain a significant portion of the variance in *initial cost per citizen* (Adjusted $R^2 = .204$, $F_{(3,\ 34)} = 4.160$, $p = .013$) (see Table 13). However, of the three categories, only *project constituents* exert a significantly positive impact on *initial cost per citizen* ($\beta = .472$, $t = 3.156$, $p = .003$); *organizational constituents* ($\beta = .132$, $t = 0.882$, $p = .384$) and *technological constituents* ($\beta = .043$, $t = 0.294$, $p = .771$) have no effect on the initial cost per citizen (see Table 13). Hypothesis 4 is thus supported whereas Hypotheses 2 and 3 are unsupported. In other words, the greater the number of project constituents, the higher the initial cost per citizen disclosed for a municipality's e-government project.

3.3.1. Post hoc analysis

Though we have demonstrated that project constituents drive the cost estimates of technological investments in municipalities' e-government projects, it could be the case that a few select case elements or even a single case element is responsible for cost predictions. Post hoc analysis was therefore performed to examine whether an individual IT business case element can significantly impact the initial cost per citizen. Again, multicollinearity was not a threat to our regression model. From Table 13, it is clear that none of the business case elements has a significant effect on *initial cost per citizen*. That is, the initial cost estimate per citizen for the municipalities' e-government projects is not correlated with the presence or absence of discrete IT business case elements. Furthermore, the regression model does not explain a significant portion of the variance in *initial cost per citizen* (Adjusted $R^2 = .155$, $F_{(8,\ 29)} = 1.846$, $p = .109$) (see Table 4).

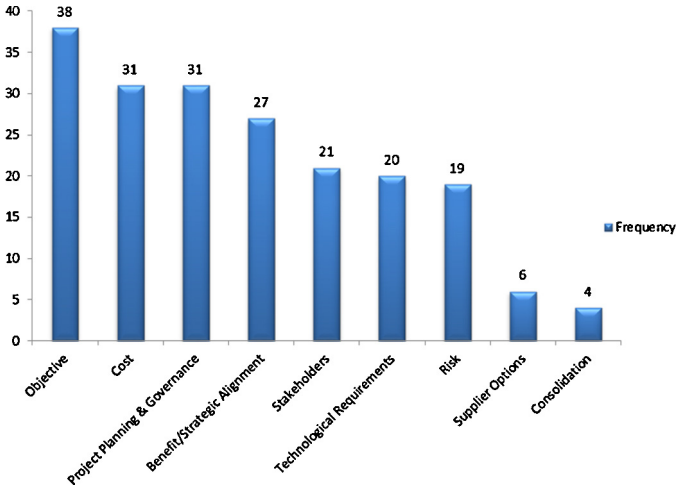


Fig. 2. Presence of business case elements for e-government project across municipalities.

4. Discussion

This study argues that business cases are an indispensable tool for improving the success of IT investments. Because managers are typically constrained by the amount of resources they can devote to the evaluation of IT investments, there is value to be gained from an in-depth appreciation of the relevance of business cases in driving the cost estimates of such investments. A well-developed business case for an IT project conserves resources in two ways. First, it channels resources to the evaluation of core elements that are critical to project success. Second, it brings to light key indicators that alert managers to economically unviable and/or failing projects.

Synthesizing the extant literature, we uncovered nine constituent elements of business cases that are pertinent to the evaluation of IT investments. Building on the work of Yetton et al. [86], these elements were then grouped into three theoretically distinct categories (i.e., *organizational constituents*, *technological constituents* and *project constituents*), each reflecting a higher-order consideration in IT project management. When combined, the nine constituent elements together with their respective higher-order categories denote an integrative framework to guide the development of business cases for IT projects. We then construct a theoretical model that postulates the impact of these business case elements on the initial cost estimates of IT investments. An empirical validation of our theoretical model based on a content analysis of e-government business cases from 38 Dutch municipalities bears implications for both theory and practice.

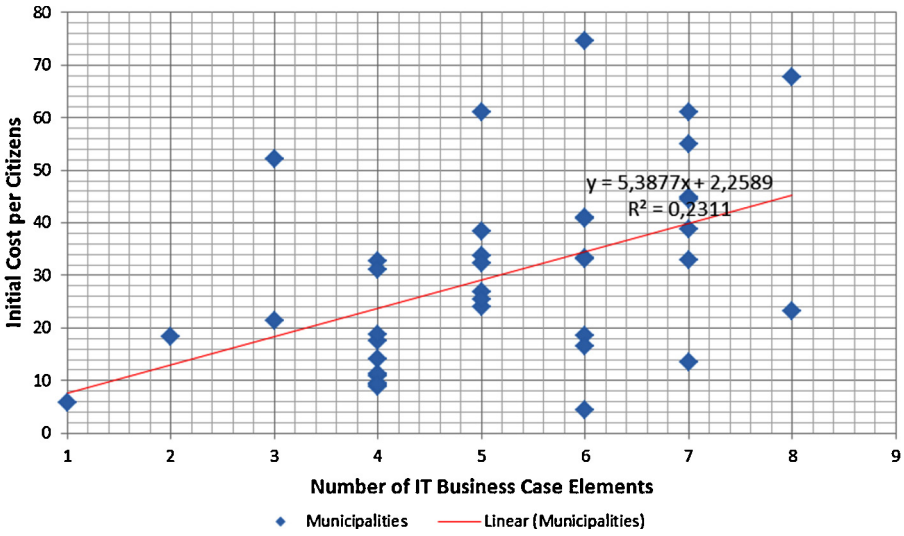


Fig. 3. Plot of initial cost per citizen against number of IT business case elements.

4.1. Implications for theory and practice

Our content analysis reveals that seven of the nine constituent elements in our proposed developmental framework can be found in the e-government business cases of at least half (or 50%) of the 38 municipalities. These elements include '*Business Case Objectives*', '*Cost Appraisal*', '*Project Planning and Governance*', '*Benefits Appraisal*', '*Stakeholders*', '*Technological Requirements*' and '*Risk Assessment*'. Our study is one of the first to corroborate previous scholarly claims, in that these seven business case elements truly determine of whether an IT investment may be worthwhile (e.g., [2,4,12,20,30,62,65,66,75,84]). As our developmental framework reflects the majority of case elements elicited from the e-government business cases (see Figure 2), we can infer that it is both rich and robust in capturing aspects of IT projects that are crucial to their success.

'*Supplier Options*' is found in only 6 of the e-government business cases, as many municipalities choose to delegate their e-government projects to 'tried' suppliers. Although such suppliers may prove to be more trustworthy in their ability to deliver on project requirements, a failure to broaden one's search for alternate suppliers represents a missed opportunity to re-evaluate existing vendor–client arrangements and identify avenues for reducing investment costs [4,7,8]. Moreover, '*Consolidation*' is found in 4 of the e-government business cases. While the inclusion of this element could be construed a repetitive and redundant summary of the other eight elements, this is not the case. '*Consolidation*' is a conclusive overview of an IT project developed by obtaining stakeholders' formal agreement on essential business case elements such as benefits, costs and milestones. '*Consolidation*' hence secures stakeholders' commitments to an IT project and forms the basis for its evaluation upon completion [81]. For the aforementioned reasons, we would caution against the exclusion of '*Supplier Options*' and '*Consolidation*' despite their absence from most of the e-government business cases.

Our empirical findings indicate a significant, positive relationship between the number of business case elements and the '*Initial Cost per Citizen*' for the e-government projects. This implies that the inclusion of additional business case elements from our developmental framework translates to improved cost estimates for IT projects. Conceivably, it is advisable to develop detailed business cases because they will safeguard against over-optimism on the part of decision makers and provide a more precise picture of the costs to be incurred (see also [23,83]), both of which have been cited by Nelson [53] as leading causes of failed IT projects.

We further discovered that different categories of business case elements have distinct impacts on the initial cost estimates of IT projects. Of the three categories (i.e., *organizational constituents*, *technological constituents* and *project constituents*), business case elements associated with project governance (i.e., '*Project Planning and Governance*', '*Cost Appraisal*', '*Risk*' and '*Stakeholders*') exert the strongest influence on the initial cost estimates. Conversely, organizational and technological business case elements do not affect initial cost estimates. Such an observation, while unexpected, is not entirely surprising. Based on meta-review of 99 IT projects, Nelson [53] advanced a series of 'best practices' for IT projects that coincide with business case elements associated with project governance: (1) avoid poor cost estimation; (2) avoid ineffective stakeholder management; (3) avoid insufficient risk management; and (4) avoid inadequate planning and quality assurance.

From the post hoc analysis, we demonstrate that none of the nine business case elements impacts the initial cost estimates of IT projects on its own. This indicates that an optimal mix of business case elements is required to arrive at improved initial cost estimates for IT projects. That is, our findings hint at the possibility of the existence of synergistic effects among business case elements: the collective consideration of certain elements yields improved initial cost estimates for IT projects. Though '*Project Planning and Governance*', '*Cost Appraisal*', '*Risk Assessment*' and '*Stakeholders*' have little influence on initial cost estimates separately, their combination in a business case provides more realistic cost estimates for e-government projects. For instance, the recognition of the various stakeholders involved in an IT investment is less informative than the identification of the risks associated with their involvement.

Finally, our investigation corroborates prior claims regarding the value of the socio-technical approach in that strict financial evaluations of IT projects could be insufficient (e.g., [24]). Financial evaluations are likely helpful as constraints for IT projects; illustrating the extent to which the project is affordable. To the best of our knowledge, this study is the first to supply empirical evidence that not only testifies to the usefulness of developing business cases for IT projects, but also alludes to the importance of rich business cases in identifying project costs.

4.2. Limitations

There are several limitations to this study. Despite extensive efforts to locate existing literature on development methodologies for IT business cases, we acknowledge that our proposed developmental framework is neither definitive nor exhaustive. However, as the nine business case elements in our developmental framework have received broad representation in past studies, we are convinced that the inclusion of these elements should constitute a solid business case for IT projects. Moreover, our content analysis fails to uncover new business case elements not already embodied in our framework.

Furthermore, it is possible that certain business case elements in our proposed developmental framework are covered in internal documents that are circulated within the municipality and but are not publicly accessible. That is, our content analysis may not fully reflect how the nine business case elements were addressed in the e-government projects of certain municipalities. Additionally, the analysis of e-government projects would reduce the generalizability of our empirical findings. Conclusions based on this investigation must therefore be interpreted with caution in that our findings would most likely be applicable to IT projects with similar characteristics.

As seen in Table 13, the business case elements in our proposed developmental framework explain a fraction of the variance in initial cost estimates of the e-government projects. We thus suspect that other factors could have influenced these estimates. For example, the experience and expertise of the business case developer may have affected his/her judgment during cost estimation. Future studies might wish to incorporate other contextual or developer-specific variables that may be pertinent to the evaluation of IT projects.

4.3. Recommendations

There are a few promising lines of research suggested by this study. First, additional analyses involving the longitudinal effects of business cases on IT projects may be warranted. Because our dataset is a cross-sectional sample of business cases from comparable e-government projects, it offers common grounds for analyzing the decisional value of business case elements. Nevertheless, the dataset does not include longitudinal data pertaining to the performance of these e-government projects. We recommend that future research conduct a post-mortem evaluation of the e-government projects to compare and contrast the extent to which the richness of business cases contributes to discrepancies between initial cost estimates and final project costs.

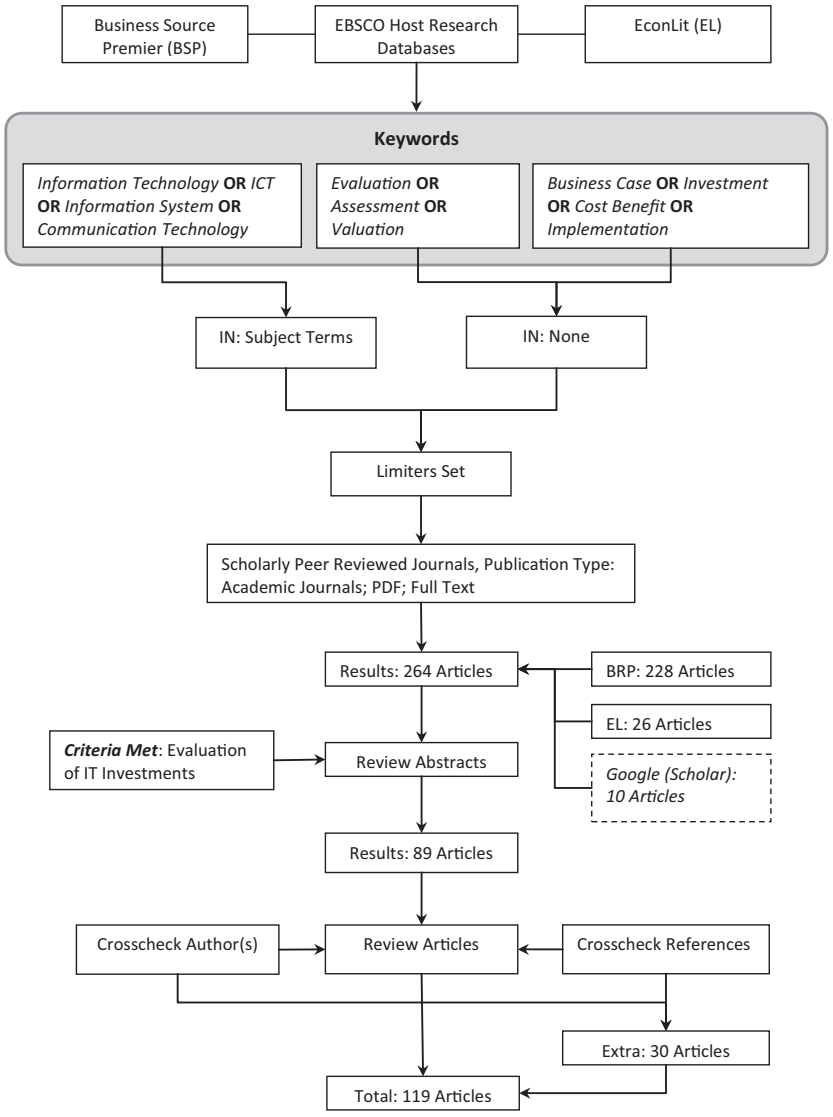
Furthermore, it would be valuable to verify whether the business case elements embodied in our proposed developmental framework can be applied to IT projects in areas other than e-government services. Applying the developmental framework beyond e-government business cases can serve as a basis for its further validation and refinement. Consequently, exploring the extent to which the developmental framework can be leveraged to develop business cases for IT projects with characteristics that are distinguishable from e-government services.constitutes a worthwhile future endeavor.

Finally, this study indicates that the richness of a business case stems from having a structured approach to the evaluation of an IT project. A well-developed business case can therefore act as a platform for scoping an IT project and configuring its requirements such that it enhances the likelihood of project success [33,45,65]. Future studies can build on our findings by examining different configurations of the nine business case elements to decipher the synergistic effects that exist among them.

4.4. Conclusion

In summary, this study makes a strong case for the development of business cases as a precursor to IT project implementation. Our research not only provides a greater appreciation for the role of business cases in IT projects, but also indicates the benefits to be gleaned from the development of rich business cases. In other words, while the development of rich business cases consume substantial resources, organizations venturing into unfamiliar IT projects will be rewarded in kind from a thorough understanding of the business worthiness of these projects. Additionally, we offer valuable lessons for both current and future e-government projects: projects with incomplete business cases or questionable cost estimations should reconsider a review of their business cases or face the risk of running into problems further down the road. Although there are ample studies on economic approaches to the evaluation of IT investment, socio-technical approaches to such evaluations remain few and far between in the extant literature, especially with respect to the growing prominence of business cases. We are not aware of any previous empirical investigations of the impact on business case methodologies on the evaluation of IT investments beyond a single case study. This paper therefore offers unique empirical evidence attesting to the importance of developing rich business cases that are consistent with our proposed developmental framework of constituent elements when undertaking IT investments.

Appendix A. Schematic overview of literature review for business case methodologies



Appendix B. Summary of extant literature on evaluation of IT projects

Author(s)	Context	Theoretical frame of reference	Conceptual vs. empirical	Evaluation criteria
Ashurst et al. [1]	IT development projects in organizations	Resource-based view of the firm	• Case study of 25 IT projects	• On-budget delivery of technology solution • On-time delivery of technology solution
Baars et al. [3]	RIFD implementation projects in organizations	N.A.	• Case studies of 3 organizations – food manufacturer, automotive and retail distribution chain	• Automation direct operational benefit – automated data capturing in goods receipt for the reduction of goods handling activities • Information direct operational benefit – automated sorting for the purpose of error reduction • Information indirect operational benefit – tracking reusable packaging to reduce loss • Information indirect managerial benefit – exception handling for unforeseen shortages or out-of-stock-situations • Transformation indirect operational benefit – using real-time data to enable a concentration of decisions at the GDC • Transformation indirect managerial benefit – analyzing root causes of spoilage with the intention of deriving organizational changes
Bechor et al. [5]	Strategic Information Systems Planning (SISP) in organizations	Contingency theory	• Survey of 172 American CIOs	• Degree of improvement in capabilities associated with the SISP process • Degree of Effectiveness in meeting objectives of the SISP process
Benlian [7]	Adoption of office suites selection	N.A.	• Online survey of 254 IS decision makers in companies	• Functionality – refers to features that the software performs and how well they meet the user's needs and requirements • Reliability – refers to the capability of the software to maintain its level of performance under stated conditions for a stated period of time • Cost – refers to expenditure associated with an application system and includes product, license, subscription, training, maintenance, and conversion costs • Ease of use – degree to which a prospective user expects the target system to be free of effort • Ease of implementation – degree of changes in policies and procedures required to implement new applications • Support – availability of maintenance and resolution options from software vendor
Benlian and Hess [8]	Evaluation of proprietary and open-source enterprise application software selection	N.A.	• Survey of 358 companies – 182 companies for proprietary software and 176 companies for open-source software	• Cost – cost involved in software acquisition, implementation and maintenance • Functionality – extent to which the software application contains all the features and functions derived from the user's business requirements • Reliability – refers to overall quality and stability of the software • Ease of use – refers to the intuitiveness of the software's user interface • Ease of customization – refers to the extent to which this package can be easily modified to meet your organization's unique needs • Ease of implementation – refers to the set-up time and effort that will be required to implement this package in your organization • Support – refers to the support (i.e. services such as training and consulting) that is provided by the vendor or OS community that has developed the software
Berente et al. [9]	Enterprise systems implementation at national aeronautics and space administration	Foucault's art of distributions	• Case study of NASA's agency-wide implementation of an ERP system in 2004 – interviews with 22 finance and administration personnel and manager, 15 ERP implementation team leaders and members, 23 center managers, research managers, and researchers, and 8 project managers	• Control principle of distributions – locate individuals for supervision • Principle of organization of geneses – delimit and structure processes • Composition of forces – depersonalize individual identities • Control of activity – constrain actions of individuals
Bose et al. [13]	RIFD implementation projects in organizations	N.A.	• Conceptual	• Value creation in terms of improved efficiency and effectiveness in operations • Value measurement in terms of rendering benefits from RFID more measurable
Conboy [16]	Large systems development in multinationals	Control systems theory	• Case study of four system development projects – interviews with 17 project managers	• Financial performance – meeting budgetary target
Davis et al. [18]	Enterprise-level system implementation	N.A.	• Survey of 75 companies	• User satisfaction with implemented system

Grimsley and Meehan [26]	E-government information systems design	N.A.	• Case study of home connections (HC) – an early e-Government project that introduced information and communication technologies (ICT) into the allocation of public and social housing in central London, U.K. • Survey-based study of clients from a similar project, Thurrock Choice Homes (TCH)	• Well-informedness when using the e-government system • Personal control when using the e-government system • Influence when using the e-government system • Satisfaction of using the e-government system • Trust in using the e-government system
Häkkinen and Hilmola [28]	ERP evaluation during the shakedown phase	DeLone and McLean's (2003) IS success model	• Case study on the effects of ERP implementation on a company's after-sales division	• Information quality – reliability/accuracy, currency, completeness and usefulness • Service quality – ease of learning, response time and reliability, system flexibility, integration on systems, usefulness of system • System quality – training, general support material, personal user support, answering to requests for change, user authorizations • Net benefits – productivity, quality of work, time to make decisions, decision quality, service to external interest groups, service to internal interest groups
Henry et al. [29]	Development of IS project in organizations	N.A.	• Survey of 216 IT professionals with project management responsibilities	• Meeting schedule and cost estimates
Kesseler [35]	Evaluation of Commercial Off-the-Shelf (COTS) software solution	N.A.	• Content study performed on the relevant certification documents with respect to the use of COTS software • Comparison between COTS-based and a carefully selected custom made case	• Improved responsiveness • Reduced time-to-market • Lower costs
Koh et al. [36]	ERP implementation in supply chains	N.A.	• Case study of ERP vendors or consultants from 10 global firms	• Holistic enterprise – real holistic "360 degree" picture of the extended enterprise, Provides ability to focus on core competencies and have a clearer picture of constraints to minimize weaknesses • Real-time information – accelerated data flow, improved accuracy of data flow and hence increased response rate (in real-time) to critical data, real-time view of partners –suppliers/ customers • Decision making power – faster, more reliable decision making based on a single source of truth, More leverage from a firm's ERP system – more "AI than hired intelligence" • Operational cost reduction – drive additional benefits through into the business and drive out further costs by additional operational cost reductions on top of traditional ERP capabilities, Assumes a firm's supply chain operations are already lean and efficient • Supply/demand efficiencies – minimal inventory and more accurate supply to customers, More planning and control "pull-like" system capabilities • Value creation – provides more ways to manage the value stream, to simplify it, and to create more value in a shorter space of time, Additional value creation of core competencies can lead to competitive advantage • Lean process optimization – integrated SC enables more process optimization improvements – mainly due to higher level of information/data, To drive more business process efficiency and further reduce costs • Core competency integration – linking core business processes internally and externally to make them more effective and efficient • Customer demand responsiveness – ability to respond more quickly and effectively to customer needs and demands, Faster response time leads to better long-term customer relationships • Improved product innovation – improved view of customers enables product innovation opportunities, to further satisfy customer needs and enhance value-creating activities, partner collaboration presents opportunities for enhanced product customization by incorporating the full SC, rather than focusing on development independently
Law and Ngai [38]	ERP system adoption	N.A.	• Case study of three ERP-using companies – IT executives were interviewed and project documents were collected from them and reviewed	• User satisfaction with ERP system

(*Continued*)

Author(s)	Context	Theoretical frame of reference	Conceptual vs. empirical	Evaluation criteria
Li and Mao [40]	CRM system evaluation	Behavioral control and motivation theories	• Case study of Nodes, a small Chinese IT firm founded in 1996 – interviews with 8 informants, including the general manager, the vice general manager, 2 sales managers and 4 sales representatives, who were all CRM users	• Enhancing outcome measurability – performance metrics were decomposed into interim results and Performance was measured more precisely, timely, and flexibly • Serving as the carrier of newly institutionalized sales processes – sales processes were configured into CRM in advance enhancing sales process visibility, activities could be monitored in real time and behaviors could be observed more completely • Serving as the carrier of the new institutional sales philosophy – the embedded "best practices" set a benchmark for the sales philosophy and Facilitating information sharing and team coordination • Serving as a more effective self-control tool
Lim et al. [42]	Evaluation of electronic tax filing system	Doney et al.'s (1998) Typology of trust-building process and Sheppard and Sherman's (1998) characterization of relational grammars	• Case study of the Inland Revenue Authority of Singapore – interviews with 89 informants, including the CIO, the assistant director of IT service division, system engineers, the corporate communication manager, customer service representatives and taxpayers	• Calculative-based trust – extent to which the trustor believes that the cost incurred by the trustee for deceitful behaviors outweigh the benefits • Capability-based trust – extent to which the trustor believes that the trustee possesses the abilities to fulfill his or her promises • Prediction-based trust – extent to which the trustor is able to predict future outcomes based on the current actions of the trustee • Transference-based trust – Extent to which the trustor has assurances from neutral third parties attesting to the trustworthiness of the trustee • Intentionality-based trust – Extent to which the trustor believes the actions of the trustee to be altruistic and benevolent
McNaughton et al. [47]	IT service management in organizations	Information technology infrastructure library framework	• Design research approach to create the framework and a contextual inquiry of industry experts was used to assess it	• Management perspective: that of highest level and/or executive management in the organization, included to ensure that areas such as financial impact, business impact, and the collective user experience are properly considered in the evaluation • Technology perspective: that of IT/IS management and the IT department as a whole. It ensures a positive effect on areas such as technological efficiency, IT personnel and resource efficiency, and specific IT related costs and budgets • IT users' perspective: that of the individuals who need the systems, technology, equipment, products and services of the IT department on a daily basis to support business processes. Users also interact with the IT department via means such as the IT help or service desk. This is included to ensure that the effect on areas such as IT service quality, expectations, and perceptions are considered as it relates to their work • IT employees' Perspective: that of the personnel within the IT department, especially operational staff affected by ITIL-related change. This may include first and second level support staff, network administrators, security personnel, database administrators, and application owners; they are the workers who directly address the implementation of new processes and changes. This perspective serves as a check on the validity of the IT user perspective and vice versa; these perspectives should be comparable
Mithas et al. [49]	Evaluating IT value at the firm level	Resource-based perspective of the firm	• Secondary data analysis of 452 firms for which complete data on key variables of interest were available from 1998 to 2003	• Profitability – net income (in millions of dollars) per employee
Nevo and Wade [54]	Evaluating firm level benefits of IT-enabled resources	Systems theory and resource-based view of the firm	• Survey of 168 firms that are members of the International Customer Service Association	• Synergy between an IT asset and an Organizational resource
Ravishankar et al. [60]	Evaluation of knowledge management system	Strategic IS Alignment	• Case study of ITS, an India-based global information technology (IT) services and consulting company – Interviews with 60 informants including software developers, middle-level managers, senior manager, research analysts and top management	• Business-IT alignment
Seddon et al. [67]	Evaluating enterprise system implementation	N.A.	• Tests of the model were conducted using data from 126 customer presentations from SAP's 2003 and 2005 Sapphire U.S. conferences	• Functional fit – extent to which the functional capabilities embedded and configured within an ES package match the functionality that the organization needs to operate effectively and efficiently
Siau and Rossi [68]	Evaluation of modeling methods for system analysis	N.A.	• Comparative analysis of different modeling methods	• Feature comparison

Srivastava and Teo [69]	Information system development	Control theory perspective	• Field study comprising 160 ISD projects executed by Indian vendors	• Quality performance – refers to the achievement of specified quality standards for the developed software • Cost performance – relates to appropriate cost savings and financial performance realized for the client through the systems development project
Taylor et al. [73]	Evaluation of IT project	Contingency perspective	• Case study of a project management office	• Project management risk – project manager experience, project management maturity • Relational risk – key stakeholder involvement, size of project group • Solution ambiguity risk – complexity and uncertainty, number of connecting systems • Environment risk – criticality, cost and duration
Thiesse et al. [74]	Integrated RIFD implementation projects in organizations	N.A.	• Case study of Kaufhof retail store	• Lead times • Labor cost reduction • Data quality • Inventory record accuracies
Wamba and Chatfield [78]	RFID implementation projects in manufacturing and logistics environment	Contingency perspective and diffusion of innovation theory	• Case study of a seven-layer TPL supply chain in Canada – Focused on logistics activities involved in managing telecommunications stationary batteries	• Organizational coordination and control – enables real-time data collection and sharing among the supply chain stakeholders • Technological capabilities for product information storage and tracking
Wang [79]	Evaluation of E-commerce system success	DeLone and McLean's (2003) IS success model	• Survey of 240 users of e-commerce systems in Taiwan	• Information quality – reliability/accuracy, currency, completeness and usefulness • Service quality – ease of learning, response time and reliability, system flexibility, integration on systems, usefulness of system • System quality – training, general support material, personal user support, answering to requests for change, user authorizations • User satisfaction in using the system • Perceived value in using the system • Intention to reuse the system
Wu et al. [85]	Evaluation of IT projects	Principal–agent formalisms	• Analytical model of optimal contract design with principal–agent techniques	• Reduction of project risks • Learning benefits • Increase development effort

References

[1] C. Ashurst, N.F. Doherty, J. Peppard, Improving the impact of IT development projects: the benefits realization capability model, European Journal of Information Systems 7 (4), 2008, pp. 352–370.

[2] Australian Government, The ICT business case guide development and review (public version), Internet document, 2008, Available via: http://www.finance.gov.au/budget/ict-investment-framework/docs/ICT_Business_Case_Guide.pdf (accessed 30.05.12).

[3] H. Baars, D. Gille, J. Struker, Evaluation of RFID applications for logistics: a framework for identifying, forecasting and assessing benefits, European Journal of Information Systems 18 (6), 2009, pp. 578–591.

[4] R. Barnes, Good Buys in IT: A Manager's Guide to Effective Spending, McGraw-Hill, London, 1995.

[5] T. Bechor, S. Neumann, M. Zviran, C. Glezer, A contingency model for estimating success of strategic information systems planning, Information & Management 47 (1), 2010, pp. 17–29.

[6] S.A. Becker, M.L. Bostelman, Aligning strategic and project measurement systems, IEEE Software 6 (3), 1999, pp. 46–51.

[7] A. Benlian, Is traditional, open-source, or on-demand first choice? Developing an AHP-based framework for comparison of difference software models in office suites selection European Journal of Information Systems 20 (5), 2011, pp. 542–559.

[8] A. Benlian, T. Hess, Comparing the relative importance of evaluation criteria in proprietary and open-source enterprise application software selection – a conjoint study of ERP and Office systems, Information Systems Journal 1 (6), 2011, pp. 503–525.

[9] N. Berente, U. Gal, Y. Yoo, Dressage, control, and enterprise systems: the case of NASA's full cost initiative, European Journal of Information Systems 19 (1), 2010, pp. 21–34.

[10] F. Bergerona, L. Raymond, S. Rivard, Ideal patterns of strategic alignment and business performance, Information & Management 41 (8), 2004, pp. 1003–1020.

[11] E.W. Berghout, T. Remenyi, Ten years of the European conference on IT evaluation: retrospectives and perspectives for possible future research, Electronic Journal of Information Systems Evaluation 8 (2), 2005, pp. 81–142.

[12] D. Borenstein, B.P. Betencourt, A multi-criteria model for the justification of IT investments, Information Systems and Operational Research 3 (1), 2005, pp. 1–22.

[13] I. Bose, E.W.T. Ngai, T.S.H. Teo, S. Spiekermann, Managing RFID projects in organizations, European Journal of Information Systems 8 (6), 2009, pp. 534–540.

[14] E.T. Brigham, P.R. Daves, Intermediate Financial Management, Dryden Press, Chicago, 2003.

[15] T.A. Byrd, B. Lewis, R. Bryan, The leveraging influence of strategic alignment on IT investment: an empirical examination, Information & Management 43 (3), 2006, pp. 308–321.

[16] K. Conboy, Project failure en masse: a study of loose budgetary control, in ISD projects, European, Journal of Information Systems 19 (3), 2010, pp. 273–287.

[17] A. Cresswell, M. LaVigne, S. Simon, S. Dawes, D. Connelly, S. Nath, J. Ruda, And justice for all: designing your business case for integrating justice information, Center for Technology in Government, University of Albany, New York, 2000, Available via: http://www.ctg.albany.edu/publications/guides/and_justice_for_all/and_justice_for_all.pdf (accessed 30.05.12).

[18] J.M. Davis, W.J. Kettinger, D.G. Kunev, When users are IT experts too: the effects of joint IT competence and partnership on satisfaction with enterprise-level systems implementation, European Journal of Information Systems 8 (1), 2009, pp. 26–37.

[19] K. El Emam, A.G. Koru, A replicated survey of IT software project failures, IEEE Software 5 (5), 2008, pp. 84–90.

[20] B. Farbey, F. Land, D. Targett, How to Assess Your IT Investment: A Study of Methods and Practice, Butterworth-Heinemann, Oxford, 1993.

[21] B. Farbey, F. Land, D. Targett, Moving IS evaluation forward: learning themes and research issues, Journal of Strategic Information Systems 8 (2), 1999, pp. 189–207.

[22] G. Fitzgerald, Evaluating information systems projects: a multidimensional approach, Journal of Information Technology 3 (1), 1998, pp. 15–27.

[23] F. Fortune, D. White, Framing of project critical success factors by a systems model, International Journal of Project Management 4 (1), 2006, pp. 53–65.

[24] F. Frisk, From business case to value case – assessing the organizational value of IT investments, in: Proceedings of the 17th European Conference on Information Systems, Verona, Italy, June 8–10, 2009, Available via: http://www.ecis2009.it/papers/ecis2009-0469.pdf (accessed 07.11.11).

[25] I. Gambles, Making the Business Case: Proposals that Succeed for Projects that Work, Gower Publishing, Farnham, 2009.

[26] M. Grimsley, A. Meehan, e-Government information systems: evaluation-led design for public value and client trust, European Journal of Information Systems 6 (2), 2007, pp. 134–148.

[27] J.F. Hair, R.E. Anderson, R.L. Tatham, W.C. Black, Multivariate Data Analysis, Prentice Hall, Englewood Cliffs, NJ, 1998.

[28] L. Hakkinen, O-P. Himola, ERP evaluation during the shakedown phase: Lessons from an after-sales division, Information Systems Journal 8 (1), 2008, pp. 73–100.

[29] R.M. Henry, G.E. McCray, R.L. Purvis, T.L. Roberts, Exploiting organizational knowledge in developing IS project cost and schedule estimates: an empirical study, Information & Management 44 (6), 2007, pp. 598–612.

[30] L. Hitt, E. Brynjolfsson, Productivity, profit and consumer welfare: three different measures of information technology's value, MIS Quarterly 20 (2), 1996, pp. 121–142.

[31] J.J. Jiang, G. Klein, R. Discenza, Information system success as impacted by risks and development strategies, IEEE Transactions on Engineering Management 8 (1), 2001, pp. 46–55.

[32] L.A. Kappelman, R. McKeeman, L. Zhang, Early warning signs of IT project failure: the dominant dozen, Information Systems Management 3 (4), 2006, pp. 31–36.
[33] A.I. Katz, Measuring technology's business value, Information Systems Management 0 (1), 1993, pp. 33–39.
[34] M. Keil, J. Mann, A. Rai, Why software projects escalate: an empirical analysis and test of four theoretical models, MIS Quarterly 4 (4), 2000, pp. 631–664.
[35] E. Kesseler, Assessing COTS software in a certifiable safety-critical domain, Information Systems Journal 8 (3), 2008, pp. 299–324.
[36] S.C.L. Koh, A. Gunasekaran, T. Goodman, Drivers, barriers and critical success factors for ERPII implementation in supply chains: a critical analysis, Journal of Strategic Information Systems 20 (4), 2011, pp. 385–402.
[37] E.W. Larson, D.H. Gobeli, Significance of project management structure on development success, IEEE Transactions on Engineering Management 6 (2), 1989, pp. 119–125.
[38] C.C.H. Law, E.W.T. Ngai, ERP systems adoption: an exploratory study of organizational factors and impacts of ERP success, Information & Management 44 (4), 2007, pp. 418–432.
[39] C.S. Lee, Modeling the business value of information technology, Information & Management 39 (3), 2001, pp. 191–210.
[40] L. Li, J.-Y. Mao, The effect of CRM use on internal sales management control: an alternative mechanism to realize CRM benefits, Information & Management 49 (6), 2012, pp. 269–277.
[41] H. Li, J. Sun, Principal component case-based reasoning ensemble for business failure prediction, Information & Management 48 (6), 2011, pp. 220–227.
[42] E.T.K. Lim, C.W. Tan, D. Cyr, S.L. Pan, B. Xiao, Advancing public trust relationships in electronic government: the Singapore e-filing journey, Information Systems Research 3 (4), 2012, pp. 1110–1130.
[43] P.E.D. Love, Z. Irani, A. Ghoneim, M. Themistocleous, An exploratory study of indirect ICT costs using the structured case method, International Journal of Information Management 6 (2), 2006, pp. 167–177.
[45] A. Martin, What drives the configuration of information technology projects? Exploratory research in 10 organizations Journal of Information Technology 8 (1), 2003, pp. 1–15.
[46] J. Matejka, T. Cossé, The Business Case Method: An Introduction, Robert F. Dame, Inc, 1981.
[47] B. McNaughton, P. Ray, L. Lewis, Designing an evaluation framework for IT service management, Information & Management 47 (4), 2010, pp. 219–225.
[48] N. Melville, K. Kraemer, V. Gurbaxani, Information technology and organizational performance: an integrative model of IT business value, MIS Quarterly 8 (2), 2004, pp. 283–321.
[49] S. Mithas, A. Tafti, I. Bardhan, J.M. Goh, Information technology and firm profitability: mechanisms and empirical evidence, MIS Quarterly 36 (1), 2012, pp. 205–224.
[50] A. Money, A. Twite, D. Remenyi, Effective Measurement and Management of IT Costs and Benefits, Butterworth-Heinemann, Newton, MA, 1995.
[51] R. Montealegre, M. Keil, De-escalating information technology projects: lessons from the Denver international airport, MIS Quarterly 4 (3), 2000, pp. 417–447.
[52] P. Neirotti, E. Paolucci, Assessing the strategic value of information technology: an analysis on the insurance sector, Information & Management 44 (6), 2007, pp. 568–582.
[53] R. Nelson, IT project management: infamous failures, classic mistakes and best practices, MIS Quarterly Executive 6 (2), 2007, pp. 67–78.
[54] S. Nevo, M. Wade, Firm-level benefits of IT-enabled resources: a conceptual extension and empirical assessment 20 (4), 2011, pp. 403–418.
[55] OECD, The Impact of the Crisis on ICTs and their Role in the Recovery, 2009, [Available via: http://www.oecd.org/dataoecd/33/20/43404360.pdf (accessed 07.11.11).
[56] K.-M. Osei-Bryson, M. Ko, Exploring the relationship between information technology investments and firm performance using regression splines analysis, Information & Management 42 (1), 2004, pp. 1–13.
[57] M.M. Parker, R.J. Benson, H.E. Trainor, Information Economics: Linking Business Performance to Information Technology, Prentice-Hall International, Englewood Cliffs, NJ, 1988.
[58] J. Peppard, R. Lampert, C. Edwards, Whose job is it anyway? Organizational information competencies for value creation Information Systems Journal 10 (4), 2000, pp. 291–322.
[59] J. Procaccino, J. Verner, S. Lorenzet, Defining and contributing to software development success, Communications of the ACM 9 (8), 2006, pp. 79–83.
[60] M.N. Ravishankar, S.L. Pan, D.E. Leidner, Examining the strategic alignment and implementation success of a KMS: a subculture-based multilevel analysis, Information Systems Research 2 (1), 2011, pp. 39–59.
[61] B. Reich, I. Benbasat, Factors that influence the social dimensions of alignment between business and information technology objectives, MIS Quarterly 4 (1), 2000, pp. 81–113.
[62] D. Remenyi, I.T. Investment, Making a Business Case, Oxford, Butterworth-Heinemann, 1999.
[63] T.J.W. Renkema, The IT Value Quest: How to Capture the Business Value of IT-based Infrastructure, J. Wiley & Sons, Chichester, 2000.
[64] T.J.W. Renkema, E.W. Berghout, Methodologies for information systems investment evaluation at the proposal stage: a comparative review, Information and Software Technology 39 (1), 1997, pp. 1–13.
[65] M. Schmidt, What's a Business Case? And Other Frequently Asked Questions Solution Matrix, Ltd., Boston, MA, 2003.
[66] Scottish Government, Scottish Capital Investment Manual: Business Case Guide, 2009, Available via: http://www.scim.scot.nhs.uk/PDFs/Manuals/BC/BC_Guide_-Full.pdf (accessed 30.05.12).
[67] P.B. Seddon, C. Calvert, S. Yang, A multi-project model of key factors affecting organizational benefits from enterprise systems, MIS Quarterly 4 (2), 2010, pp. 305–328.
[68] K. Siau, M. Rossi, Evaluation techniques for systems analysis and design modeling methods – a review and comparative analysis, Information Systems Journal 1 (3), 2011, pp. 249–268.
[69] S.C. Srivastava, T.S.H. Teo, Contract performance in offshore systems development: role of control of mechanisms, Journal of Management Information Systems 9 (1), 2012, pp. 115–158.
[70] Standish Group, The CHAOS Summary 2009 Report, The Standish Group International, Boston, MA, 2009.
[71] R. Stockdale, C. Standing, An interpretive approach to evaluating information systems: a content, context, process framework, European Journal of Operational Research 172 (3), 2006, pp. 1090–1102.
[72] A. Taudes, M. Feurstein, A. Mild, Options analysis of software platform decisions: a case study, MIS Quarterly 4 (2), 2000, pp. 227–243.
[73] H. Taylor, E. Artman, J.P. Woelfer, Information technology project risk management: bridging the gap between research and practice, Journal of Information Technology 7 (1), 2012, pp. 17–34.
[74] F. Thiesse, J. Al-Kassab, E. Fleisch, Understanding the value of integrated RFID systems: a case study from apparel retail, European Journal of Information Systems 8 (6), 2009, pp. 592–614.
[75] J.T.M. van der Zee, B. de Jong, Alignment is not enough: integrating business and information technology management with the balanced business scorecard, Journal of Management Information Systems 6 (2), 1999, pp. 137–156.
[76] W. van Grembergen, Information Technology Evaluation Methods and Management, Idea Group Publishing, Hershey, PA, 2001.
[77] G. Walsham, Interpretive evaluation design for information systems, in: L.P. Willcocks, S. Lester (Eds.), Beyond the Productivity Paradox, John Wiley & Sons Ltd., London, 1999.
[78] S.F. Wamba, A.T. Chatfield, A contingency model for creating value from RFID supply chain network projects in logistics and manufacturing environment, European Journal of Information Systems 8 (6), 2009, pp. 615–636.
[79] Y-S. Wang, Assessing e-commerce systems success: a respecification and validation of the Delone and McLean model of IS success, Information Systems Journal 8 (5), 2008, pp. 529–557.
[80] J.L. Ward, Recognizing Project Warning Signs, 2003, Available via: http://www,esi-intl.com/public/publications/122003executivewarningsigns.asp (accessed 07.11.11).
[81] J. Ward, E. Daniel, J. Peppard, Building better business cases for IT investments, MIS Quarterly Executive 7 (1), 2008, pp. 1–15.
[82] J. Ward, S. De Hertog, S. Viaene, Managing benefits from IS/IT investments: an empirical investigation into current practice, in: Proceedings of the 40th Hawaii International Conference on Systems Science, Hawaii, USA, January 3–6, 2007.
[83] B. Whittaker, What went wrong? Unsuccessful information technology projects Information Management & Computer Security 7 (1), 1999, pp. 23–30.
[84] L. Willcocks, Information Management: The Evaluation of Information Systems Investments, Chapman and Hall, London, 1994.
[85] D.J. Wu, M. Ding, L.M. Hitt, IT implementation contract design: analytical and experimental investigation of IT value, learning and contract structure, Information Systems Research 2012http://dx.doi.org/10.1287/isre.1120.044, forthcoming.
[86] P. Yetton, A. Martin, R. Sharma, K. Johnston, A model of information systems development project performance, Information Systems Journal 10 (4), 2000, pp. 263–289.



Prof. Dr. Egon Berghout is Chair in Information Systems at the University of Groningen and independent Board Advisor. He is also president of the Benelux Chapter of the Association for Information Systems (AIS). He is specialised in the creation of sustainable strategic advantage through IT. He received his PhD from Delft University of Technology, an Executive Master's degree in IT Auditing from Erasmus University and MSc (Information Management) from Tilburg University.



Chee-Wee Tan is an Associate Professor at Copenhagen Business School. He received his PhD from the University of British Columbia, Canada. His research interests focus on design, innovation and evaluation issues related to technological services. Findings from his research has been presented at numerous international conferences and he has articles published or forthcoming in journals such as *MIS Quarterly* (MISQ), *Information Systems Research* (ISR), *Journal of the American Society for Information Science and Technology* (JASIST), *Information & Management* (I&M), *European Journal of Information Systems* (EJIS) and *Decision Support Systems* (DSS) among others.